Mark schemes

Q1.

(a) Sample mean = 53.06, s = 1.140

Both. For s AWRT 1.14

B1

t_5 = 2.571

AWRT 2.57

B1

Sample mean ±t × s / √6

For √6

M1

Rest of formula. Allow t_5 = 2.01 to 2.02, or t_6 = 2.45

m1

(53.06 ± 1.20) = (51.86 , 54.26)

Either form ± 0.01 in total.

A1

5

(b) Sample mean is lower than last year's mean so claim **may** be true.
53.41 lies within c.i. so **not certain** that mean time is better.
Performance in competition does not depend on mean time.
Times seem to be improving.

E1 each for sensible comments either supporting or against statement up to a maximum of 2.
Comment must be ***uncertain****.*

E2

2

[7]

Q2.

(a) (i) $\bar{x} = \dfrac{2290}{50}$ = 45.8 or 45800

CAO

B1

$(s^2 =)\ \dfrac{28225.5}{49 \text{ or } 50}$ **or** $(s =)\ \sqrt{\dfrac{28225.5}{49 \text{ or } 50}}$

Ignore notation

M1

s = 24(.0) or 24000 to 24001

AWRT / AWFW *(24.00064)*

(σ = 23.75942)

SCs: (for no seen working)
M1 A1 for 24.0 or 24000 to 24001
M1 A0 for 24 or 23700 to 23800

A1

3

(ii) $\bar{x} - ns$ = (45.8 − n × 24.0) < 0
SC: Accept quoted values of
(−4 to −1) (n = 2) or (−28.5 to −23.5) (n = 3) (both AWFW)

Allow (45 to 47) and any multiple of
(23.5 to 24.5) which gives value < 0

Must clearly state the value of a numerical expression

M1

and negative salaries are impossible

OE; must be in context

Negative values impossible $\Rightarrow$ *A0*

A1

2

Alternative solution 1

P(X < 0 | N(45.8, 24.0^2) = P(Z < −1.91)

Standardising 0 using 45.8 & 24.0

(M1)

= 0.027 to 0.03

In addition to probability within range,
must state that negative salaries are impossible

(A1)

(2)

Alternative solution 2

P(X > 60 | N(45.8, 24.0^2)) = P(Z > 0.59)

Standardising 60 using 45.8 & 24.0

(M1)

= 0.27 to 0.28

In addition to probability within range,
must compare calculated value to 6 / 50 = 0.12 OE

(A1)

(2)

(b) 99% (0.99) ⇒ z = 2.57 to 2.58

AWFW *(2.5758)*

B1

CI for μ is $\bar{x} \pm z \times \frac{s}{\sqrt{n}}$

Used with ($\bar{x}$ & s) from (a)(i) and
z(1.64 to 2.58) & ÷ $\sqrt{n}$ with n > 1

M1

Thus $45.8 \pm 2.5758 \times \frac{24.0}{\sqrt{50}}$

F on ($\bar{x}$ & s) with ÷ $\sqrt{50 \text{ or } 49}$ &
z(1.64 to 1.65 or 2.32 to 2.33 or 2.57 to 2.58)

AF1

Hence 45.8 ± (8.7 to 8.8)
or 45800 ± (8700 to 8800)
OR
(37.(0) to 37.1, 54.5 to 54.6)
or (37000 to 37100, 54500 to 54600)

CAO / AWFW *(8.74)*
Ignore (absence of) quoted units
AWFW

A1

4

(c) See supplementary sheet for additional illustrations

Clear correct comparison of 55 or 55000 with c's UCL or CI

Accept 55000 compared with
c's 54.5 to 54.6 (ie different units)

B1

(6/50 or 0.12 or 12%) </≠ 0.25 or 25%

OE; correct comparison mentioning both 12% and 25%

B1

Reject both / each of the two claims

Dependent on B1 B1

Bdep1

3

Additional comment illustrations

It / (claimed) mean / (claimed) value > UCL / CI

Must indicate 55 or 55000

(B0)

99% have (mean) weights between CLs so...

(B0)

Any comparison of 60 (£60 000) with UCL / IC

Value of 60 does not refer to mean

(B0)

P(X < 60 | N(45.8, 24.0^2)) = P(Z > 0.59)

= (0.27 to 0.28) < 6 / 50 = 0.12

Assumes salaries ~ N; cf (a)(ii)

(B0)

[12]

Q3.

(a) (i) Mean, $\bar{x} = 20.4$

CAO

B1 **1**

(ii) Standard deviation $s = \sqrt{\dfrac{400.24}{64}}$

Expression must be seen $\left(\sqrt{6.25375}\right)$

M1

= 2.50

AWRT (2.50075)

A1 **2**

NB: $s = \dfrac{400.24}{64 \text{ or } 65}$ (6.15754 or 6.25375)

No $\sqrt{\ }$ *and / or use of divisor n*

(B1)

or

$s = \dfrac{\sqrt{400.24}}{65}$ (2.48144)

Use of divisor n

(B1)

(b) 96%(0.96) $\Rightarrow$ z = 2.05 to 2.06

AWFW (2.0537)

B1

CI for μ is $\bar{x} \pm z \times \dfrac{s}{\sqrt{n}}$

Used

Must have $\sqrt{n}$ *with n > 1*

M1

Thus 20± 2.0537 × $\dfrac{2.50}{\sqrt{65 \text{ or } 64}}$

F on $\bar{x}$ *and z*

A1F

Hence 20.4 ± 0.6 **or** (19.8, 21(.0))

AWRT

A1 4

(c) CI in (b)(i) contains/includes 20

Or equivalent
Dependent on CI in (b)(i)

B1F

thus

no (significant) evidence to support claim

Or equivalent
Dependent on B1F

Bdep1 2

[9]

Q4.

(a) (i) $\dfrac{0.98+1.00}{2}$ or $\dfrac{0.975+1.005}{2}$ or

$0.98+\dfrac{0.02}{2}$ or $0.975+\dfrac{0.03}{2} = 0.99$

AG
(At least) one correct expression seen

Ignore contradictions
Accept any valid equivalent

B1

(ii) $\dfrac{0.97+0.98}{2} = 0.975$ and $\dfrac{1.00+1.01}{2} = 1.005$

Both CAO
Can not be implied from (a)(i)

B1

2

SC In (a)(i) and (a)(ii) allow $1.004\dot{9}$ or 1.0049… etc
Similar forms for lower boundary

(b) Mean, $\bar{x}$ = 1.062

CAO $\sum fx$ *=106.2* *Ignore notation*

B1

Standard deviation, s or σ = 0.043

AWRT $\sum fx^2 = 112.9662$

If B0 B0, M1 can be awarded for attempt at $\dfrac{\sum fx}{100}$

B2

3

(c) (i) 99%(0.99) $\Rightarrow$ z = 2.57 to 2.58

AWFW (2.5758)

B1

$t_{99}(0.995) = 2.626$ *AWRT*

(B1)

CI for μ is $\bar{x}$ ± (z or t) × $\dfrac{(s \text{ or } \sigma)}{\sqrt{n}}$

Used
Must have √n with n > 1

M1

Thus 1.062 ± 2.5758 × $\dfrac{0.043}{\sqrt{100 \text{ or } 99}}$

F on $\bar{x}$*, s/σ and z/t*

A1F

Hence 1.06 ± 0.01 or (1.05, 1.07)

AWRT; award even if previous inaccuracies in $\bar{x}$, s/ σ or z/t
Dependent on A1F

A1

4

(ii) Sample data grouped

σ unknown

Exact sample values unknown / midpoints used

s calculated from a sample

$\bar{x}$ and *s* calculated from grouped data

$\bar{x}$ (not μ) and s are estimates
NOT data values rounded

B1

1

(d) (i) CI for *μ* or CI in (c)(i) > 1
LCL of CI for *μ* or
LCL of CI in (c)(i) > 1

Or equivalent; must compare CI to 1
Dependent on CI in (c)(i) > 1

B1

(ii) 99 or 100 or all sample/ table/ data volumes/ values/ *x*-values/ cartons are within this range
(or none / 0 or 1 volumes outside)

B1

2

[13]

Q5.

(a) (i) 96% (0.96) $\Rightarrow$ z = **2.05 to 2.06**

AWFW *(2.0537)*

B1

CI for *μ* is $\bar{x} \pm z \times \frac{s}{\sqrt{n}}$

Used with 251.1 and 1.94 correctly

Must have $\sqrt{n}$ with n > 1

M1

Thus $251.1 \pm 2.0537 \times \frac{1.94}{\sqrt{50 \text{ or } 49}}$

F on z only

AF1

Hence **251.1 ± 0.6**
or **(250.5, 251.7)**

CAO/AWRT

Dependent on AF1 but not on z so can be gained using an incorrect z

AWRT

Adep1

4

(ii) Claim is $\mu > 250$

Clear correct comparison of **250** with **LCL or CI**

F on CI (250 < LCL or CI)

BF1

so
Claim is supported/reasonable/correct/true/etc
Must be consistent with c's comparison

Dependent on BF1

Bdep1

2

(b) (i) I for x is $\bar{x} \pm z \times s$
= 251.1 ± 2.0537 × 1.94

Must have n = 1 and correct or same z as in (a)(i)

M1

Hence **251.1 ± 4(.0)**

CAO/AWRT

or

(247, 255)

F on z in (a)(i); can be gained using an incorrect z

AWRT

AF1

2

(ii) Some individual packets are likely to/will contain less than 250 grams

Or equivalent F on (b)(i)

BF1

1

[9]

Q6.

(a) $\sqrt{\frac{184.5}{49}}$ **or** $1.92 \times \sqrt{\frac{50}{49}}$

Fully correct** expression or equivalent must be **seen

Note: $s = \sqrt{184.5/50} = 1.939 \Rightarrow$ B0

B1

= 1.94

AG

1

(b) (i) 96% (0.96) $\Rightarrow$ z = **2.05 to 2.06**

AWFW *(2.0537)*

B1

CI for μ is $\bar{x} \pm z \times \frac{s}{\sqrt{n}}$

Used with 251.1 and 1.94 correctly

Must have $\sqrt{n}$ *with n > 1*

M1

Thus $251.1 \pm 2.0537 \times \frac{1.94}{\sqrt{50 \text{ or } 49}}$

F on z only

AF1

Hence **251.1 ± 0.6**

or

(250.5, 251.7)

CAO/AWRT

Dependent on AF1 but not on z so can be gained using an incorrect z

AWRT

Adep1

4

(ii) Claim is μ > 250

Clear correct comparison of **250** with **LCL or CI**

F on CI *(250 < LCL or CI)*

BF1

so

Claim is supported/reasonable/correct/true/etc

Must be consistent with c's comparison

Dependent on BF1

Bdep1

2

(c) $\bar{x} - ns = 251.1 - n \times 1.94 < 250$

SC: Quoted values of 249.2, 247.2 or 245.3 (AWRT) $\Rightarrow$ M1

Allow any multiple of 1.94

*Must **clearly indicate** the value of a numerical expression giving a result less than 250*

M1

so
Some individual packets are likely to/will contain less than 250 grams

Or equivalent

A1

2

[9]

Q7.

(a) 60 containers are a **random sample**
OR are **selected independently**

OE; eg representative

B1

99% $\Rightarrow$ z = 2.57 to 2.58

AWFW (2.5758)

B1

OR
99% $\Rightarrow$ t = 2.66(0)

CAO

(B1)

(Knowledge of the t-distribution is not required in this unit)

CI for μ is $\bar{x} \pm (z \text{ or } t) \times \frac{s}{\sqrt{n}}$

Used; must have $\sqrt{n}$ *with n > 1*

M1

Hence

10.191 ± (2.5758 or 2.66) × $\frac{0.13759}{\sqrt{60}}$

ft on z or t only

A1ft

Hence 10.191 ± (0.0456 to 0.0473)

Hence 10.19 ± (0.04 to 0.05)
OR
(10.14 to 10.15, 10.23 to 10.24)
AWFW

A1
5

(b) Value of 10 is **below / outside** CI
ft on (a)(i); OE

B1ft

Suggests **mean volume** is
greater than / **more than** / **different from** 10 litres
ft on (a)(i); OE

B1ftdep

OR

$\bar{x} - 2s = 9.91582$

(B1)

Suggests some **volumes** may be **less than** 10 litres

(B1dep)
2

[7]

Q8.

$n = 100$ $\bar{x} = 1.90$ $s = 3.32$

(a) Mean = 60 + $\bar{x}$

M1

= 61.9
CAO

A1

Standard deviation = 3.32
CAO

B1
3

(b) 98% $\Rightarrow$ $z = 2.32$ to 2.33
AWFW (2.3263)

($\Rightarrow$ t = 2.36 to 2.37)

AWFW (2.364)

B1

CI for μ is $\bar{x} \pm z / t \times \frac{s}{\sqrt{n}}$

Used; must have $\sqrt{n}$ *with n > 1*

M1

Thus 61.9 ± 2.3263 × $\frac{3.32}{\sqrt{100}}$

ft on (a) and z/t only

A1ft

Hence 61.9 ± (0.7 to 0.8)

or (61.1 to 61.2, 62.6 to 62.7)

Accept 1.03 ± (0.012 to 0.013)
AWFW
Accept (1.01 to 1.02, 1.04 to 1.05)

A1

4

(c) $\bar{S}$ *>> 1 hour or 60 minutes:*

Not valid as UCL = 1 hour

(Accept Both limits = 1 hour)

Dependent on
UCL = 62.6 to 62.7 or
UCL = 1.04 to 1.05

B1dep

1

[8]

Q9.

(a) (i)

x:	–5	–3	–1	1	3	5	7	9
F:	4	9	13	27	21	15	7	4

Mean $(\bar{x}) = 1.9$

CAO (190)

B2

(0.9 to 2.9)

AWFW

(B1)

Standard deviation (s_{n-1} or σ_n) =

(1452)

3.3(0) to 3.32

AWFW (3.31967)

B2

(3(.00) to 3.5(0))

AWFW (3.30303)

(B1)

If no marks scored but $\sum fx$ attempted and result divided by 100

(M1)

4

(ii) Mean = 60 + $\bar{x}$

M1

= 61.9

ft on (a)(i)

A1ft

Standard deviation = 3.3(0) to 3.32

ft on (a)(i); accept 'same as' only providing answer in (a)(i)

B1ft

3

(b) (i) 98% $\Rightarrow$ z = 2.32 to 2.33

AWFW (2.3263)

B1

($\Rightarrow$ t = 2.36 to 2.37)

AWFW (2.364)

CI for μ is $\bar{x} \pm z / t \times \dfrac{s_{n-1} \text{ or } \sigma_n}{\sqrt{n \text{ or } n-1}}$

Used; must have $\sqrt{n}$ *with n > 1*

M1

Thus $61.9 \pm 2.3263 \times \dfrac{3.3 \text{ to } 3.32}{\sqrt{100 \text{ or } 99}}$

ft on (a)(ii) and z/t only

A1ft

Hence 61.9 ± (0.7 to 0.8)
or (61.1 to 61.2, 62.6 to 62.7)

Accept 1.03 ± (0.012 to 0.013)
AWFW
Accept (1.01 to 1.02, 1.04 to 1.05)

A1

4

(ii) Mean and SD based upon grouped data
SD (**not mean**) calculated from a sample
CLT used / Times (may) not (be) normal

Actual times/values unknown
Or equivalent

B1

1

(c) *S > 1 hour or 60 minutes:*
Valid as 74/100 or 0.74 or 74% > 50%

Must use 74 etc
Or equivalent

B1

$\bar{S}$ >> 1 hour or 60 minutes:
Not valid as UCL ≈ 1 hour
(Accept Both limits ≈ 1 hour)

Dependent on
UCL = 62.6 to 62.7 or
UCL = 1.04 to 1.05

B1dep

2

[14]

Q10.

(a) 90% $\Rightarrow$ z = 1.64 to 1.65

AWFW (1.6449)

B1

or
90% $\Rightarrow$ t = 1.66 to 1.67

AWFW (1.6649)

(B1)

(Knowledge of the *t*-distribution is **not** required in this unit)

CI for μ is $\bar{x} \pm (z \text{ or } t) \times \frac{(s_{n-1} \text{ or } s_n)}{\sqrt{n}}$

Used; must have $\sqrt{n}$ *with n > 1*

M1

Thus

$184 \pm (1.6449 \text{ or } 1.6649) \times \frac{(32 \text{ or } 32.2)}{(\sqrt{78} \text{ or } \sqrt{77})}$

ft on z or t only

A1ft

Hence 184 ± (5.94 to 6.13)
or £184 ± £6
or (£178, £190)

AWRT; ignore units

A1

4

(b) (i) **Likely** to be **valid**

Accept 'valid' or equivalent

B1

(ii) Different plays have different:
programme prices, sales, marketing, etc
theatre or audience sizes, etc
popularity, artists, etc

B1

so

↑Dep↑

Unlikely to be **valid**

Accept 'not valid' or equivalent

B1

3

[7]

Q11.

(a) 95% $\Rightarrow$ z = 1.96

CAO

B1

or
95% $\Rightarrow$ t = 2.0 to 2.01

AWFW (2.009)

(B1)

(Knowledge of the t-distribution is not required in this unit)

CI for μ is $\bar{x} \pm (z \text{ or } t) \times \frac{(s_{n-1} \text{ or } s_n)}{\sqrt{n}}$

Used; must have $\sqrt{n}$ *with n > 1*

M1

Note that $25.1 \times \sqrt{\frac{50}{49}} = 25.35483$

$25.1 \times \frac{50}{49} = 25.61224$

Max of B1 M1 A0ft A1

Thus

$234 \pm (1.96 \text{ or } 2.009) \times \frac{(25.1 \text{ or } 25.3 \text{ to } 25.4)}{(\sqrt{50} \text{ or } \sqrt{49})}$

ft on z or t only

A1ft

Hence 234 ± (6.95 to 7.30)

ie 234 ± 7
or (227, 241)

AWRT

A1

4

(b) Customers are likely to choose large / similar sized potatoes

OE; accept any sensible alternative

B1

1

[5]

Q12.

(a) Mean = $\frac{286.5}{50}$ = 5.73

CAO

B1

Standard deviation = $\sqrt{\frac{45.16}{49 \text{ or } 50}}$ = 0.95 to 0.961

AWFW

B1 **2**

(b) 99% $\Rightarrow$ z = 2.57 to 2.58

AWFW *2.5758*

B1

CI for μ is $\bar{x} \pm z \times \frac{(\sigma \text{ or } s)}{\sqrt{n}}$

Use of

Must have $\div\left(\sqrt{n}\right)$ *with n > 1*

M1

Thus 5.73 ± 2.5758 × $\frac{(0.95 \text{ to } 0.961)}{\sqrt{50}}$

ft on z and $S^2/\bar{x}$ *> 0 but not on x Accept only 50 or 49 for n*

A1ft

5.73 ± (0.34 to 0.36)

↑

(5.37 to 5.39, 6.07 to 6.09)

AWFW

A1 **4**

(c) CI excludes both values of 5 and $6\frac{1}{2}$

ft on(b); OE

B1ft

so

↑

Neither claim appears valid

Dependent

ft on(b); OE

B1ft

or

CI excludes 5 so claim not valid

ft on(b); OE

(B1ft)

and

CI excludes $6\frac{1}{2}$ so claim not valid
ft on(b); OE

(B1ft)
2
[8]

Q13.

(a) Weight, $X \sim N(1012, 5^2)$

(i) $P(X < 1015) = P\left(Z < \frac{1015 - 1012}{5}\right)$

standardising (1014.5, 1015 or 1015.5) with 1012 and ($\sqrt{5}$, 5 or 5^2) and/or (1012 – x)

M1

$= P(Z < 0.6)$
CAO; ignore sign

A1

$= \Phi(0.6) = 0.725$ to 0.726
AWFW *(0.72575)*

A1
3

(ii) $P(X > 1005) = P\left(Z > \frac{1005 - 1012}{5}\right)$

standardising (1004.5, 1005 or 1005.5) with 1012 and ($\sqrt{5}$, 5 or 5^2) and/or (1012 – x)

M1

$= P(Z > -1.4) = P(Z < 1.4)$
area change

m1

$= 0.919$ to 0.920

AWRT *(0.91924)*

A1

3

(iii) P(1005 < X < 1015) = (i) – [1 – (ii)]

or equivalent

M1

= 0.72575 – [1 – 0.91924]
= 0.644 to 0.646

AWFW *(0.64499)*

ft on (i) and (ii) providing > 0

A1ft

2

(b) (i) Mean, $\bar{y}$ = 502.85

CAO; stated or implied

B1

Variance, σ^2 / s^2 = 50.3 to 51.5
or
Standard deviation, σ / s = 7.09 to 7.17

AWFW; either stated or implied

B1

98% $\Rightarrow$ z = 2.32 to 2.33

AWFW *(2.3263)*

B1

or
98% $\Rightarrow$ t = 2.40

B1

(Knowledge of the t-distribution is not required in this unit)

AWRT *(2.403)*

B1

CI for μ is $\bar{y} \pm (z \text{ or } t) \times \dfrac{(\sigma \text{ or } s)}{\sqrt{n}}$

use of; must have $\left(\div \sqrt{n}\right)$ *with n > 1*

M1

Thus

$$502.85 \pm (2.33 \text{ or } 2.40) \times \left(\frac{7.09 \text{ to } 7.17}{\sqrt{49 \text{ or } 50}}\right)$$

ft on $\bar{x}$, z/t and σ/s > 0;
n = 49 or 50 only

A1ft

or
502.85 ± (2.32 to 2.46)
or
(500.3 to 500.6, 505.1 to 505.4)
AWFW; dependent on A1

A1 dep

6

(ii) **Given sample:**
6 in 50/some of packets have weights below 500 grams
or equivalent

B1

Confidence interval:
CI > 500
ft on CI in (b)(i)

B1ft

Conclusion:
Statement does not appear justified
or equivalent.
dependent on both B1 and B1ft

B1 dep

3

[17]

Q14.

(a) (i) Time, $X \sim N(12, 2.5^2)$

P (X < 15) = P $\left(Z < \frac{15 - 12}{2.5}\right)$
standardising (14.5, 15 or 15.5) with
($\sqrt{2.5}$, 2.5 or 2.5^2) and/or (12 – x)

M1

P (Z < 1.2) = 0.885
AWRT (0.88493)

A1

2

(ii) P (10 < X <15) = (i) – P (X < 10)

OE

M1

= 0.88493 – P (Z < 0.8)
= 0.88493 – (1Φ (0.8))

area change

= 0.88493 – (1 – 0.78814) = 0.673

AWRT (0.67307)

A1

3

(b) (i) $\bar{y} = \frac{835.0}{50} = 16.7$

CAO

B1

$s^2 = \frac{533.61}{49}$ = 10.89 or s = 3.3

CAO; either

$v = \frac{533.61}{50}$ = 10.6722 or $\sqrt{v}$ = 3.2668

AWRT 10.67 or AWRT 3.27

B1

99% $\Rightarrow$ z = 2.5758

AWFW 2.57 to 2.58

B1

CI for μ is

$\bar{y} \pm z \times \frac{(s \text{ or } \sqrt{v})}{\sqrt{n}}$

use of; must have (÷ $\sqrt{n}$) with n >1

M1

Thus: 16.7 ± 2.5758 × $\frac{(3.3 \text{ or } 3.27)}{\sqrt{50}}$

ft on $\bar{y}, z, (s \text{ or } \sqrt{v})$; not on n

A1ft

Thus: (15.5, 17.9)

AWRT; dependent on ÷ 49 for variance unless subsequently corrected

A1

6

(ii) Adding 25% to 12 gives 15

CAO; seen somewhere

B1

Since 15 is outside/below CI

ft on (b)(i); must use 15

E1ft

Mustafa's suspicion is supported

ft on (b)(i); must use 15

B1ft

3

[14]

Q15.

(a) Var(T) = $s^2 = \frac{279.8929}{49}$ = 5.71

awrt (5.7121)

B1

SE($\overline{T}$) = $\sqrt{\frac{\text{Var}(T)}{50}}$

use of

M1

= 0.338

*Awrt [**cannot** be scored in part (b)(i)]*

A1

3

(b) (i) $\bar{t} = \frac{143.5}{50}$ = 2.87

cao; can be scored in part (a)

B1

99% implies z = 2.5758

awfw 2.57 to 2.58

B1

CI for μ is: $\bar{t} \pm z \times \frac{(s \text{ or } \sigma)}{\sqrt{n}}$

use of; must have $\sqrt{n}$ with n > 1 or equivalent

M1

or $\bar{t} \pm z \times \text{SE}(\bar{t})$

or $\sqrt{n}$ in SE($\bar{t}$)

Thus: 2.87 ± (2.5758 × 0.338)

f.t. on $\bar{t}$, z and SE ($\bar{t}$) > 0; accept
$\bar{t}$ = 143.5 only if clearly stated

A1f.t.

Thus: (2.00, 3.74)

awrt; accept 2
dependent on ÷ by 49 in part (a) unless subsequently corrected

A1

5

(ii) Evidence to suggest that μ = 3.5

f.t. on part (b)(i)

B1f.t.

as 3.5 inside CI

clearly stated; f.t. on part (b)(i)

B1f.t.

2

(c) Now evidence (to suggest that μ has changed/increased from 3.5 (as 3.5 outside/below CI)

reason not required

B1

Also evidence (to suggest μ has increased during three months) as CIs do not overlap

reason not required

B1

2

[12]

Q16.

(a) $$\hat{\mu} = \bar{x} = \frac{1}{n}\sum x = \frac{1040}{100} = 10.4$$

CAO

B1

$\hat{\sigma}^2 = s^2 = \frac{1}{n-1}\left(\sum x^2 - \frac{\left(\sum x\right)^2}{n}\right)$

use of, or use of $\frac{n}{n-1}$ *v or v*

M1

$= \frac{1}{99}\left(11102.11 - \frac{1040^2}{100}\right) = 2.89$

CAO (v = 2.8611) ($\sqrt{v}$ *= 1.69148)*

A1

3

(b) CI : $\bar{x} \pm z \times \frac{s}{\sqrt{n}}$

Use of with n > 1

M1

99% $\Rightarrow$ *z* = 2.5758

AWFW 2.57 to 2.58

B1

$\therefore$ 10.4 ± 2.5758 × $\frac{1.7}{\sqrt{100}}$

A1ft

$\therefore$ 10.4 ± 0.44

ft on (a) providing $\bar{x}$ *≠ 1040, & on z, not on n*

A1

i.e. (9.96, 10.8)

AWRT; dependent on ÷ by 99 in part (a)
unless subsequently corrected

A1dep

4

(c) Require to subtract 0.2 from each CL

subtract/add 0.2 from/to each CL

M1

$\therefore$ (9.76, 10.6)

ft on (b); AWRT

A1ft

2

[9]

Q17.

(a) Unbiased estimate of σ^2 is given by

$$\frac{n \times (\text{sample standard deviation})^2}{n-1}$$

use of; accept omission of ^{2}

M1

$$= \frac{100 \times 2.97^2}{99} \text{ or } \frac{100 \times 8.82(09)}{99}$$

CAO; OE

A1 2

= 8.91

AG

(b) 99% $\Rightarrow$ z = 2.5758

AWFW 2.57 to 2.58

B1

CI for μ is $\bar{x} \pm z \times \frac{(s \text{ or } \sigma)}{\sqrt{n}}$

use of; must have $\sqrt{n}$ *with n > 1*

M1

$\therefore$ 331.28 ± 2.5758 × $\frac{\sqrt{8.91}}{\sqrt{100}}$

ft on z only

A1f.t.

$\therefore$ 331.28 ± 0.77

$\therefore$ (330.51, 332.05)

AWRT (dep on A1f.t.)

A1 4

[6]

Q18.

(a) $\sum x$ = 25065 $\quad \sum (x - \bar{x})^2 = 384.16$

Mean, $\bar{x} = \frac{25\,065}{50}$ = 501.3

CAO

B1

Variance, $s^2 = \frac{384.16}{49}$ = 7.84 or s = 2.8

CAO

B1

[v = 7.68 or $\sqrt{v}$ = 277]

AWRT

99% $\Rightarrow$ z = 2.5758

AWFW 2.57 to 2.58

B1

CI for μ is: $\bar{x} \pm z \times \frac{(s \text{ or } \sigma)}{\sqrt{n}}$

Use of ; must have $\sqrt{n}$ *with n > 1*

M1

$\therefore$ 501.3 ± 2.5758 × $\frac{(2.8 \text{ or } \sqrt{7.84})}{\sqrt{50}}$

ft on $\bar{x}$*, s and z not on n*

A1f.t.

$\therefore$ 501.3 ± 1.02

$\therefore$ (500.28, 502.32)

AWRT

A1

6

(b) Probably = 0.01

CAO

Allow (1 – (0.99 × 0.90))

B1 M1

1

(c) Probability = 0.01 × 0.10 = 0.001

CAO

A1

2

[9]

Q19.

(a) $A \sim N(10.3, 0.16^2)$

(i) $P(A < 10.5) = P\left(Z < \frac{10.5 - 10.3}{0.16}\right)$

standardising 10.45, 10.5, 10.55 with σ
σ^2, $\sqrt{\sigma}$ and /or $(\mu\text{-}x)$

M1

$= P(Z < 1.25) =$

CAO ± 1.25

A1

0.89 (435)

AWRT

A1

3

(ii) $P(A > a) = 0.75$

$\Rightarrow z = -0.6745$

AWFW (±)0.674 to ±0.675
(not 0.67 or 0.68)

B1

$\frac{a - 10.3}{0.16} = z$

standardising value with σ, σ^2, $\sqrt{\sigma}$ and/or $(\mu - x)$

M1

$\Rightarrow \frac{a - 10.3}{0.16} = -0.6745$

equating two z-values (m0 for using 1-z-value)

m1

$a = 10.2$

AWRT

A1

4

(b) (i) $\hat{\mu} = \bar{x} = 24.9$

CAO
(B0 for bias correction)

B1 1

(ii) SE($\hat{\mu}$) = $\frac{\sigma}{\sqrt{n}}$ =

use of (σ, σ^2, $\sqrt{\sigma}$) ÷ (n, $\sqrt{n}$) with $n > 1$

(accept with 0.16, 0.16^2, $\sqrt{0.16}$)

(M0 for bias correction)

M1

0.02

Cannot gain (i) and (ii) from (iii)

CAO

A1 2

(iii) 95% $\Rightarrow$ z = 1.96

CAO

B1

CI for μ is: $\hat{\mu} \pm z \times$ SE($\hat{\mu}$)

use of; must have used n>1 in (ii)

M1

$\therefore$ 24.9 ± 1.96 × 0.02

$\therefore$ 24.9 ± 0.04

f.t. on (i) and (ii)

A1f.t.

24.86, 24.94

AWRT

A1 4

(iv) As 95% CI excludes 25

f.t. on (iii)

B1f.t.

Disagree with claim

f.t.on (iii); dep on B1f.t.

E1f.t. 2

Only mark if CI of the form (a,b) with a and b determined

[16]

Q20.

(a) $\bar{x}$ = 4256/400 = 10.64

10.64 allow 10.6

B1

95% confidence interval for mean

1.96

B1

$10.64 \pm 1.96 \times \frac{3.68}{\sqrt{400}}$

Use of $\frac{3.68}{\sqrt{400}}$, *allow use of* $3.68\sqrt{\frac{400}{399}}$

M1

10.64 ± 0.361

Correct method for interval, their mean- allow incorrect z–value

m1

(10.28 , 11.00)

10.28 (10.275 ~ 10.3) and
11.00 (10.995 ~ 11.005)
or 10.64cao ± 0.361 (0.36 ~ 0.361)

A1

5

(b) (i) $E(X) = 5 \times 0.15 + 10 \times 0.63 + 15 \times 0.15 + 20 \times 0.07$

Method for E(X)

M1

= £10.7

10.7 cao – ignore units

A1

(ii) $E(X^2) = 25 \times 0.15 \times 100 \times 0.63 + 225 \times 0.15 + 400 \times 0.07 = 128.5$

Method for $E(X^2)$ *may be implied*

M1

Method for s.d., their answers to (i) and (ii) – allow variance if called variance

M1

(iii) s.d. of $X = \sqrt{128.5 - 10.7^2}$ = £3.74

Completely correct method for s.d. –

Allow variance if called variance

m1

3.74 (3.74 ~ 3.745)

A1

6

(c) mean within confidence interval calculated in (a), standard deviation close to observed standard deviation. Model appears plausible.

ft Mean within confidence interval –

E1ft

allow similar
ft s.d. similar to observed
Correct conclusion based on correct calculations

E1ft

3

(d) (i) $\bar{x}$ = 2342/200 = 11.71

11.71 or 11.7

B1

H_0 μ = 11.00

One correct hypothesis – generous

B1

H_1 μ > 11.00

Both correct – ungenerous

B1

$$z = \frac{11.71 - 11.00}{\frac{3.42}{\sqrt{200}}} = 2.94$$

M1

Use of $\frac{3.42}{\sqrt{200}}$ *, allow use of 3.42* $\sqrt{\frac{200}{199}}$

M1

Correct method for z, ignore sign

m1

2.94 (2.93 ~ 2.94)

A1

critical value 1.6449

1.6449 oe 1.645 or 1.64 or 1.65
allow t = 1.652 on 1.653

B1

Reject H_0 significant evidence mean exceeds

A1ft

8

Correct conclusion, their figures, must be compared with appropriate tail of z.
needs previous

M1

(ii) Since £11 was upper limit of confidence interval for mean, there is strong evidence that the mean has increased

Evidence sales have increased

E1

Since £11 upper limit of confidence interval

E1

2

(iii) Have total sales of petrol increased? How much does the scheme cost? Have other sales increased? Etc

Any sensible point

E1

A second sensible point

E1

2

[26]

Q21.

(a) $x = \frac{4256}{400} = 10.64$

10.64 allow 10

B1

95% confidence interval for mean

$10.64 \pm 1.96 \times \frac{3.68}{\sqrt{400}}$

1.96

B1

use of 3.68/√400

M1

10.64 ± 0.361

correct method for interval – their mean - allow incorrect z-value

m1

(10.28, 11.00)

10.28 (10.275 ~ 10.3) and
11.00 (10.995 ~ 11.005)
or 10.64 cao ± 0.361 (0.36 ~ 0.361)

A1

5

(b) (i) $x = \frac{2342}{200} = 11.71$

95% confidence interval for mean

$11.71 \pm 1.96 \times \frac{3.42}{\sqrt{200}}$

completely correct method

M1

11.71 ± 0.474
(11.24 , 12.18)

11.24 (11.2 ~ 11.3) and 12.18 (12.15 ~ 12.2)
***or** 11.71 (11.7 ~ 11.71) ± 0.474 (0.473 ~ 0.475)*

A1

2

(ii) Since confidence intervals for mean before and after the offer do not overlap there is strong evidence that the mean has increased

confidence intervals do not overlap

E1ft

correct conclusion based on correct calculation and reason

E1

2

(iii) Have **total** sales of petrol increased? How much does the scheme cost? Have other sales increased? Etc

Any sensible point

E1

A second sensible point

E1

2

[11]

Q22.

(a) $197 \pm 1.96 \times \frac{103}{\sqrt{90}}$

use of $\frac{103}{\sqrt{90}}$

M1

1.96 (or allow 1.987 to 1.99)

B1

197 ± 21.3

completely correct method - their z

m1

176 ~ 218

197 ± 2.13 (21.2 to 21.7) or
176(175 to 176) and 218 (218 to 219)

A1

4

(b) 42.6

42.6(42.5 to 42.6) or 43.2(43.1 to 43.3)
if t is used

B1

1

(c) $2z \times \frac{103}{\sqrt{90}} = 30$

reasonable attempt at equation containing
z - ignore omission of 2

M1

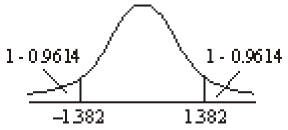
z = 1.382

m1

completely correct equation containing z

m1

method for finding z - allow omission of 2



1 – 2(1 – 0.9164) = 0.833

method for probability - their z

M1

83.3%

83.3 (83 – 83.5)

A1

5

(d) $2 \times 2.5758 \times \dfrac{103}{\sqrt{n}} = 30$

2.5758 (2.57 to 2.58)

B1

reasonable attempt at equation involving n

M1

- ignore omission of 2, incorrect z

n = 312.8

method of solution of equation

m1

313 needed

313 (310 to 315)

A1

4

[14]

Q23.

(a) $\bar{x}$ = 135.6

135.6 cao

B1

s = 0.45552

0.45552 (0.455 to 0.456) may be implied by correct interval

B1

95% confidence interval for mean is 135.6 ± 2.306 × $\dfrac{0.45552}{\sqrt{9}}$

8df

EXAM PAPERS PRACTICE

B1

2.306

B1f.t.

Use of their $\frac{s}{\sqrt{9}}$

i.e. 135.6 ± 0.350

135.25 to 135.95

Correct method for interval

m1

Answer given to 4,5 or 6 sf, must be an interval

B1

135.25 (135.2 to 135.3) and
135.95 (1359 to 136)
***Or** 135.6 **cao** ± 0.35 (0.349 to 0.351)*

A1

8

(b) 135.8 ± 1.96 × $\frac{3.9}{\sqrt{60}}$

*1.96 **cao***

B1

Use of $\frac{3.9}{\sqrt{60}}$

B1

135.8 ± 0.987

Correct method for interval

m1

134.81 to 136.79

134.81 (134.8 to 134.820) and 136.79 (136.78 to 136.8)
***Or** 135.8cao ± 0.987 (0.98 to 1)*

A1

4

(c) (i) Decrease

Decrease

B1

1

(ii) Increase - narrower interval $\rightarrow$ more likely to identify need for overhaul if $\mu \neq 135.0$

Increase

E1

Reason

E1

2

(d) Large variability as in (b) will lead to unsatisfactory production even if mean on target. Might prefer small variability as in (a) even if mean slightly off target.

Large variability unsatisfactory/makes deviation from 135 difficult to detect.

E1

Small variability with mean slightly off target may be preferable

E1

2

[17]

Q24.

(a) (i) $\bar{x} = 496.125 \qquad s = 7.318$

496.125(496 – 496.2)

B1

7.318 (7.31 – 7.33) - may be implied

B1

95% confidence interval for mean

Use of their $\frac{s}{\sqrt{8}}$ *- generous*

M1

$496.125 \pm 2.365 \times \frac{7.318}{\sqrt{8}}$

$\left.\begin{array}{r}\textit{Use of t}\\ \textit{Completely corect method}\end{array}\right\}\text{M2}$

m1

7df

B1

i.e. 496.125 ± 6.12

2.365 – their df

B1 ft.

490.0 ~ 502.2

490.0 (489.95 – 490.05) and

502.2 (502.2 – 502.3)
Allow 502 or 496.125 (496.1 – 496.2)
± 6.12 (6.1 – 6.12)

A1

8

(ii) 80% confidence interval for mean

Their t

M1

$496.125 \pm 1.415 \times \frac{7.318}{\sqrt{95}}$

i.e. 496.125 ± 3.66
492.5 ~ 499.8

492.5 (492.4 – 492.5) and
499.8 (499.7 – 499.9)
Allow 492 and 500
or 496.125 (496.1 – 496.2)
± 3.66 (3.65 – 3.67)

A1

2

(b) $499.6 \pm 1.57 \times \frac{9.3}{\sqrt{95}}$

Use of $\frac{9.3}{\sqrt{95}}$
Correct method, their z

m1

499.6 ± 1.57

498.0 – 501.2

498 (497.9 – 498.1) and
501.2 (501.1 – 501.2)
Allow 501 or 499.6 ± 1.57 (1.56 – 1.58)

A1

4

(c) (i) (A) 0.8

0.8 cao

B1

(B) 0.15

Method

M1

0.15 cao

A1

3

(ii) 0.95 × 0.2 = 0.19

Method

0.19 cao

A1

2

[19]

Q25.

(a) $\bar{x}$ = 48.4 s = 18.094

48.4 cao

18.094 (18.09 – 18.1)

B1 B1

99% confidence interval

Use of t

M1

$48.4 \pm 3.250 \times \dfrac{18.094}{\sqrt{10}}$

9 df

3.250 – their df

B1 B1ft

ie 48.4 ± 18.6

29.8 – 67.0

Use of their $\dfrac{s.d.}{\sqrt{10}}$

Completely correct method – their t

29.8(29.75 – 29.85)

and 67.0 (66.95 – 67.05)

or 48.4 and 18.6 (18.55 – 18.65)

sc B1 B1 M1 M1 available if z used

M1 m1 A1

8

(b) (i) 95% confidence interval

1.96 – allow t *1.979 – 1.984*

B1

$49.5 \pm 196 \times \dfrac{16.5}{\sqrt{110}}$

Use of $\frac{16.5}{\sqrt{110}}$

Their z

M1

49.5 ± 3.08

A1

4

46.42 – 52.58

49.5 ± 3.08(3.08 – 3.09)

or 46.42 (46.1 – 46.42)

and 52.58(52.58 – 52.59)

Allow 46.4 and 52.6

(ii) 49.5 ± 1.96 × 16.5

Their z

M1

17.2 – 81.8

49.5 ± 32.34 (32.3 – 32.4)

or 17.2 (17.1 – 17.2) and

81.8 (81.8 – 81.9)

A1

2

(c) Mean greater than 25 but many

Mean $\neq$ *25*

E1

individuals will be less than 25

Mean > 25

Some individuals < 25

E1 E1

3

(d) Interval narrower

Interval narrower

E1

No need for normal assumption

Normal not needed

***Or** better estimate of* σ*, more representative, more reliable **not** more accurate*

E1

2

(e) Both samples random so valid suggestion

valid

E1

Would be better but not much

($\frac{16.5}{\sqrt{110}}$ very similar to $\frac{16.5}{\sqrt{120}}$)

Better

E1

Not much - any three points

E1

3

[22]

Examiner reports

Q1.

The calculation part was handled well, apart from by those students who used a *z*-value, and others who ignored the instruction to give the answer to two decimal places. A fair proportion of students recognised the uncertainty introduced because the 2011 mean lay within the 2012 confidence interval, and correctly deduced that the 2012 mean might lie below or above the previous year's mean. Others were far too definite in their conclusions. Strangely, a sizeable proportion of students equated a higher time with a better performance. Very few observed that success in races is not directly linked to mean time and depends also on other athletes' performances.

Q2.

Most candidates found much of this question very challenging. In part (a)(i), it was most disappointing to see the number of candidates failing to score 3 marks. Whilst there was some excuse for candidates dividing $\sum (x - \bar{x})^2$ by 50, rather than 49, there was really no excuse for the many who did not know that $\bar{x} = \frac{\sum x}{50}$. Frequent incorrect attempts included trying to substitute $\sum x$ into $\sum(x - \bar{x})^2$ or dividing $\sum x$ by a variety of denominators. Some candidates were confused, either at the outset or later, by the units involved with invalid multiplications of s^2 by 1000 taking place at some point. In part (a)(ii), the common incorrect answers were verbose about university salaries or simply that 24.0 was too large when compared to 45.8. Even those candidates who calculated, for example (45.8 – 2 × 24.0), simply stated that a normal distribution or 'it', instead of salaries, could not take negative values. There were many partially correct answers to part (b) using follow-through answers from part (a)(i) though some candidates lost further marks for an incorrect z-value. In answering part (c), a majority of candidates attempted the comparison of 55 with their CI or UCL. However, far too many then did the same with 60 rather than compare $\frac{6}{50}$ = 12% with 25%, or considered the standardising of 60 using a normal distribution. This perhaps suggested that they had not read the question with sufficient care.

Q3.

As usual, candidates found the calculations in parts (a) and (b) easier than the explanations in part (c), with the result that about 6 marks was the norm. Almost all candidates scored the 1 mark available in part (a)(i). However, more difficulties arose in answering part (a)(ii).

All too often, candidates evaluated $\sqrt{\frac{400.24}{65}}$ and then attempted to make it unbiased by incorrectly multiplying by $\frac{65}{64}$ rather than $\sqrt{\frac{65}{64}}$. The best and simplest correct approach was $\sqrt{\frac{400.24}{64}}$. Answers to part (b) showed a sound understanding and usually scored full marks.

Q4.

Surprisingly, this was probably the worst answered question on the paper, with even the better candidates scoring no more than about 9 of the 13 marks available. Most candidates scored the mark for answering part (a)(i) but many made a meal of the calculation, rather than simply evaluating $\left(\frac{0.98 + 1.00}{2}\right)$.

Answers to part (a)(ii), which was essentially GCSE grade C material, were in the main very poor. All too often, the values quoted were 0.98 and 1.00, although numerous other incorrect values were seen. Even the better candidates, who managed to obtain the correct minimum value of 0.975, often quoted a maximum value of 1.004, 1.049, 1.0049 or even 1.05 rather than 1.005 or 1.0049.

Again in part (b), far too many incorrect answers were seen. A significant proportion of candidates used formulae, with varying success usually due to numerical or approximation errors. Even those candidates who used their calculators' inbuilt statistical functions obtained incorrect answers or quoted answers to insufficient accuracy. The most common incorrect answers were 1.065 and 0.073, obtained by using correct mid-points but ignoring frequencies.

Answers to part (c)(i) generally showed a sound understanding with at least 3 marks available despite incorrect answers in part (b). Incorrect *z*-values, omission of $\sqrt{n}$ or use of $\sqrt{8}$ were errors occasionally seen. Candidates exhibited a better understanding in answering part (c)(ii), perhaps because the idea of the use of mid-points for grouped data leading to an estimate of the mean was familiar from GCSE. Again, there was some confusion when candidates tried to discuss samples and populations, whilst some thought that the confidence interval was approximate because it was 99% rather than 100%.

Although correct answers, by comparing $\mu = 1$ with the confidence interval, were seen to part (d)(i), too many answers compared the sample mean with 1 litre or referred to 91 of the 100 cartons containing more than 1 litre. Again, in part (d)(ii), whilst some correct answers were in evidence, many candidates simply either noted that 1 litre was within the given interval or stated that 'it must contain all values as total probability is 1'. Some acceptable calculations of a normal probability were given, and a few candidates referred to $\bar{x} \pm 3s$ but rarely supported their statements with the necessary numerical justification.

Q5.

It was of concern to see the large proportion of weak answers to this question, particularly as the topic has been examined in a similar way on all previous papers. The construction of the 96% confidence interval caused problems for far too many candidates. Common errors were an incorrect *z*-value (often 1.7507); use of 251 or 250 for $\bar{x}$; use of 1.92, $\sqrt{1.94}$, 1.94^2 or even 184.5 for *s*. Those few candidates who had obtained a correct answer in part (a)(i) then struggled in part (a)(ii) by failing to clearly compare 250 (the words 'it' or 'mean' were not acceptable) with their confidence interval in order to comment on the manufacturer's claim. Some stated incorrectly that their interval contained 250 whilst others stated that the claim was false since their interval did not contain 250. Answers to part (b)(i) rarely scored marks due to candidates failing to recognise that it required a repeat of part (a)(i) with $n = 1$. As a result, only the very best candidates were able to gain the mark in part (b)(ii).

Q6.

It was most surprising and disappointing to see the large proportion of weak answers to this question, particularly as the topic has been examined in a similar way on all previous papers.

In part (a), more candidates started with $\frac{184.5}{49}$ rather than $\frac{184.5}{50}$ to obtain s^2, but some then omitted to indicate the square root as the next step. Some candidates appeared to have no idea of the steps required, and either made no attempt or tried substituting $\bar{x} = 251.1$ into $\sum(x - \bar{x})^2 = 184$.

The construction of the 96% confidence interval caused problems for far too many candidates. Common errors were an incorrect *z*-value (often 1.7507); use of 251 or 250 for $\bar{x}$; use of 1.92, $\sqrt{1.94}$, 1.94^2 or even 184.5 for *s*; divisors of 50 or even 1.

Some candidates substituted correct values into an incorrect formula, such as $z \pm \frac{s}{\sqrt{n}} \times \bar{x}$ or $\bar{x} \pm \frac{zn}{s}$.

Most of these aforementioned errors lost most, if not all, of the 4 marks. Many candidates who had obtained a correct answer in part (b)(i) then struggled in part (b)(ii) by failing to clearly compare 250 (the words 'it' or 'mean' were not acceptable) with their confidence interval in order to comment on the manufacturer's claim. Some stated incorrectly that their interval contained 250 whilst others stated that the claim was false since their interval did not contain 250. Answers to part (c) rarely scored marks due to candidates' providing a qualitative judgement rather than a numerical justification. To score the 2 marks, candidates had to show that [251.1 − (≥1) × 1.94] gave a value less than 250 and so some bags were likely to be underweight.

Q7.

Most candidates ignored the request for a necessary assumption in part (a) or stated that 'Volumes', or 'The sample' or even 'They' were normally distributed. An answer needed to indicate that the 60 containers were '*A random sample*' or were '*Selected independently*' or were '*Representative*'. Construction of the confidence interval was almost invariably correct. However teachers should note that, from page 12 of the supplied booklet, s^2 is an unbiased estimate of variance and so no 'correction' to *s* was required; but this was not penalised here.

In part (b), many candidates' answers revealed confusion between mean volume (confidence interval) and the volumes (10 litres) of individual containers. Such confusion was not helped by definitive statements.

Q8.

This question proved to be the most difficult on the paper. In part (a), many candidates made a complete hash of the deduction process by apparently not realising that 1 hour was equivalent to 60 minutes. Thus it was all too common to see candidates adding 1, or even 100, to both mean and standard deviation values or multiplying one or both values by 60 or 100. Future candidates clearly need to be much better prepared for similar questions involving linear scaling. In answering part (b), most candidates were clearly aware of the relevant formula for a confidence interval and identified that $z = 2.3263$ but lost at least 1 accuracy mark due to the aforementioned errors in part (a) or by simply using the given values of 1.90 and 3.32. The awarding of the mark in part (c) was rare indeed since the answer required a comparison of 1 hour or 60 minutes with a correspondingly **correct** confidence interval in part (b).

Q9.

This question proved to be the most difficult on the paper. It was very disappointing to see so many candidates, some scoring high marks on other questions, failing, either by using their calculators' statistical functions or from first principles using formulae, to find correct values in part (a)(i). All too often, incorrect mid-points were identified or, more seriously, frequencies were ignored. Attempts at part (a)(ii) were often equally disappointing. Whilst some candidates ignored the word 'Hence' and so repeated part (a)(i) with revised mid-points, many of those who tried to deduce answers made a complete hash of the process by apparently not realising that 1 hour was equivalent to 60 minutes. Thus it was all too common to see candidates adding 1, or even 100, to both mean and standard deviation values or multiplying one or both values by 60 or 100. Future candidates clearly need to be much better prepared in these areas of the specification. In answering part (b)(i), most candidates were aware of the relevant formula for a confidence interval but lost 1 accuracy mark due to the aforementioned errors in part (a). Candidates who used an incorrect z-value or, more seriously, used n = 8, lost more marks here. In part (b)(ii), the most common acceptable reasons were "actual times unknown" or "midpoints used" whereas the common response of "mean and standard deviation are estimates" did not gain the mark available. The scoring of marks in part (c) was rare indeed. Most candidates failed to identify the difference between Vernon's two claims and so referred to their sample mean values for both. His first claim required reference to 74% (or 26%) from the original data, whereas his second claim required a comparison of 1 hour or 60 minutes with a correspondingly **correct** confidence interval in part (b)(i). A small minority of candidates considered that the claims were best answered by commenting on Vernon's plumbing skills and work ethic.

Q10.

Most candidates scored at least 4 marks. The formula for the confidence interval was well-known and used correctly by most candidates; an incorrect z-value was the usual error. In answering part (b)(i), many comments observed that the mean for this play lay inside the confidence interval (as of course it must) and so the assumption was valid. This suggested anticipation on the part of these candidates of the type of comment that has been expected following the determination of a confidence interval on previous papers. In part (b)(ii), plays having different popularities or audience sizes were the most frequently seen acceptable reasons for the comment "Unlikely to be valid".

Q11.

A large majority of candidates scored full marks on this question. In part (a), a minority of candidates used an incorrect z-value, usually 1.6449, or, more critically, omitted to divide by $\sqrt{50}$. An 'adjustment' to the value of 25.1 for the sample standard deviation was not expected and was rarely, if ever, seen. Similarly, use of the t-distribution (Statistics 2) was very rare but not penalised when applied correctly. Almost all candidates stated a valid reason in part (b), usually referring to 'selection by size of potato'. Unacceptable reasons usually referred to a small sample size.

Q12.

Statistics 1A

As mentioned earlier, far too many candidates lost at least 1 mark in part (a). Those that did not, usually used s_n rather than s_{n-1} whereas those that did, used s^2n, s^2n-1, $\sqrt{\sum(x-\bar{x})^2}$ or even $\sum(x-\bar{x})^2$.

In part (b), the value of z was identified correctly but the divisor of $\sqrt{n}$ was omitted on too many occasions; perhaps candidates were wary after already dividing by n in part (a)? Those candidates who used 286.5 for their sample mean apparently saw nothing wrong with a clearly incorrect answer if the context was considered! The many candidates who attempted parts (a) and (b) correctly invariably provided correct answers to an appropriate level of accuracy. Those candidates who referenced their confidence interval in answering part (c) usually scored the 2 marks available. However, too many candidates merely compared the claims with their value for the sample mean or produced a qualitative argument based upon their perceived reasons as to why the fire and rescue service would underestimate whilst the parish councillor would overestimate.

Statistics 1B

As mentioned earlier, far too many candidates lost at least 1 mark in part (a). Those that did not, usually used s_n rather than s_{n-1} whereas those that did, used s^2n, s^2n-1, $\sqrt{\sum(x-\bar{x})^2}$ or even $\sum(x-\bar{x})^2$.

In part (b), the value of z was identified correctly but the divisor of $\sqrt{n}$ was omitted on too many occasions; perhaps candidates were wary after already dividing by n in part (a)? Those candidates who used 286.5 for their sample mean apparently saw nothing wrong with a clearly incorrect answer if the context was considered! The many candidates who attempted parts (a) and (b) correctly invariably provided correct answers to an appropriate level of accuracy. The many candidates who referenced their confidence interval in answering part (c) usually scored the 2 marks available. However, too many merely compared the claims with their value for the sample mean.

Q13.

Many candidates scored full marks in part (a) and it was pleasing to observe the improvements in notation and presentation. On the rare occasions when marks were lost in part (a)(i), it was for standardising 1014 or 1014.5, rather than 1015; something that needs to be eradicated for the future. In part (a)(ii), some candidates obtained the correct final answer but not always by the most direct method; many others usually failed to carry out the necessary area change. In part (a)(iii), far too many candidates assumed that all that was required was to find the difference of their two previous answers and so scored no marks. Candidates who missed the link and so began afresh, often scored full marks. Candidates need to think more carefully about such linked parts to questions.

In part (b)(i), few candidates scored all 6 marks. Despite comments in a previous Examiners Report and the relevant formula for s^2 on page 10 of the supplied booklet, far too many candidates were unable to calculate the correct value for s from the given value for $\sum(y-\bar{y})^2$; something that must be addressed for the future. Other less frequent errors were due to an incorrect value for $\bar{y}$, an incorrect z-value or omission of $\sqrt{50}$. In part (b)(ii), the awarding of full marks was even rarer. Many candidates did not make it clear that they were comparing 500 (and not the sample mean) with their confidence interval, failed to indicate that 6 in 50 packets were underweight or stated no overall conclusion.

Q14.

Most candidates were able to answer part (a) competently, often immediately spotting that the answer to part (ii) involved the answer to part (i). Weaker candidates often wanted to insert a continuity correction and a small minority had confused ideas about variance and standard deviation. In part (b)(i), confusion again arose over variance and standard deviation. Few candidates divided 533.61 by 49 immediately. Some candidates used a correction factor of $\frac{n}{n-1}$ later, but many forgot or applied it to the standard deviation. A fair sized minority of candidates did not divide by $\sqrt{50}$ when calculating their confidence interval for the mean. Most candidates correctly used z in the acceptable range of 2.57 to 2.58. Overall, few candidates scored the full six marks on this part with most scoring in the range three to five marks. A few candidates were apparently unconcerned when they found that Mustafa took upwards of 14 hours to walk to school! Only the better candidates could make a sensible attempt at answering part (b)(ii). Many candidates did not appreciate the need to add 25% onto 12 and so obtain 15, which then had to be looked at in the light of their confidence interval. Those candidates who did so, were then usually able to score all three marks.

Q15.

As mentioned previously, answers to part (a) were generally very disappointing. Most candidates making any progress divided by 50 though about a quarter subsequently scaled their answer using $\frac{50}{49}$. Candidates should be encouraged to find unbiased estimates of σ^2 using s_{n-1}^2 directly. Many candidates did use a correct expression for standard error but many omitted the square root in either the numerator or denominator.

Follow-through marks in part (b)(i) enabled many candidates to score at least four marks but the weakest candidates failed to evaluate $\bar{t}$ or find a correct z-value.

Answers to part (b)(ii) generally scored two marks for a correct deduction based on an answer to part (b)(i) though again clarity left much to be desired.

In part (c), whilst many candidates realised that the value of μ had increased/changed, only a handful of candidates referred to non-overlapping intervals for the second mark. Often comments suggested the new confidence interval was more reliable due to an increased sample size.

Q16.

As stated in the *General* comments, far too many candidates made a complete mess of answering part (a). Additionally, whilst the use of μ and σ, rather than $\bar{x}$ and s^2, was not penalised, it did perhaps indicate a lack of understanding of the difference between a population parameter and a sample statistic. Many candidates scored most of the marks in part (b) for correct follow-through working. In part (c), most candidates realised that the limits needed moving by 0.2 cm, some even indicating that this was because only the mean changed. However, all too often, 0.2 cm was added, rather than subtracted; a loss of one of the two marks.

Q17.

Most candidates scored the two marks in part (a) though, in quite a number of cases, the quoted answer appeared helpful. In part (b), candidates were aware of the general form for a confidence interval and able to obtain a z-value within the acceptable range of 2.57 to 2.58. However, many candidates lost two of the four available marks by using 2.97 rather than $\sqrt{8.91}$, whilst others lost one mark through ignoring the accuracy required.

Q18.

Answers to part (a) were often weak through candidates' inability to determine values of $\bar{x}$ and s^2 from the values of $\sum x$ and $\sum(x-\bar{x})^2$. Whilst many candidates found the value of $\bar{x}$ correctly, some simply used $\sum x$. However, it was rare indeed to see s^2 evaluated as $\frac{1}{49}\sum(x-\bar{x})^2$. Alternatives were $\frac{1}{50}\sum(x-\bar{x})^2$, rare and acceptable, or $\sum(x-\bar{x})^2$, common but not acceptable. Whilst most candidates knew the correct form for the confidence interval and identified the correct value of z for 99%, the aforementioned fundamental errors made somewhat of a nonsense of the subsequent substitutions. Consequently most accuracy marks were forfeited in part (a)(i).

Most candidates simply made no attempt at parts (b) and (c) or stated 'unlikely/zero' to both parts. Thus, in general, only the very best candidates could make worthwhile attempts and, somewhat surprisingly, many of these candidates scored two marks in part (c) but not one mark in part (b). Many candidates appeared not to have a firm grasp of the fundamental ideas of confidence intervals.

Q19.

This question was well answered by the majority of candidates. Almost without exception, all but the weakest candidates scored the 3 marks in part (a)(i) and there were few attempts at a continuity correction. In part (a)(ii), whilst method marks were often scored, candidates quoted an inaccurate z-value possibly due to use of Table 3 rather than Table 4 and/or gave an incorrect sign. Some candidates used a 'non z -value' of 1 – 0.67 (0.1) or simply 0.75. In parts (b)(i) and (ii), some candidates biased their estimate of standard error, some even of their mean, by multiplying by $\left(\frac{n}{n-1}\right)$. However, these candidates and most others then scored full marks in part (b)(iii) by ignoring such attempted corrections, though a small minority used either 2 or 1.645 rather than 1.96. Answers to part (b)(iv) were usually sufficiently clear with correct deductions from previous answers.

However, some candidates suggested that 24.94 would round to 25 or that 25 was close to the sample mean.

Q20.

Apart from candidates who thought that the mean amount spent by each of the previous 400 customers was £4256, there were few problems with parts (a) and (d)(i).

Part (b) was almost universally well answered apart from a few candidates who confused standard deviation and variance.

There were many reasonable attempts at part (d)(i) but very few candidates made the connection between the null hypothesis μ = 11 in part (d)(i) and the upper limit of the confidence interval in part (a). A variety of sensible suggestions were made in part (d)(iii)

Q21.

Apart from candidates who thought that the mean amount spent by each of the previous 400 customers was £4256, there were few problems with parts (a) and (b)(i).

In part (b)(ii) however, many candidates demonstrated that they completely misunderstood the meaning of a confidence interval. Often they did not realise that it has any connection with the mean but thought that 95% of the amounts spent on petrol would lie within the limits of the confidence interval. Many made sensible suggestions in part (b)(iii)

Q22.

Most candidates were well prepared for this question. In part (c) many candidates correctly found a z-value of 1.38 but were unable to find the percentage confidence. The answer 91% was common.

Q23.

Parts (a) and (b) were well answered. Unlike previous years, there was a mark in part (a) for giving the answer to an appropriate degree of accuracy (4, 5 or 6 significant figures were accepted). Those candidates who hedged their bets by giving alternative answers to different levels of accuracy did not receive this mark unless all their answers were to 4, 5 or 6 significant figures.

Most candidates made reasonable attempts at parts (c) and (d), although answers to part (d) revealed that many candidates did not realise that confidence intervals apply to means and are not intervals within which a given percentage of the population lies.

Q24.

The calculations posed relatively few problems, although a small minority used z where t was appropriate. There were many good answers to part (c) - a common error was to answer (c)(i)(B) as if the samples were independent, as they were in (c)(ii).

Q25.

Parts (a) and (b)(i) were well answered. The great majority of candidates successfully obtained standard deviation directly from their calculators as recommended. In part (a), a few candidates failed to use t, or used the wrong divisor for the standard deviation.

Only a minority of candidates were able to answer part (b)(ii) correctly, many merely repeated their answer to part (b)(i). Others calculated the width of the confidence interval.

The interpretation in part (c) was generally poor. Many candidates failed to make any connection between the confidence interval and the mean, believing rather that 95% of scores lie in the 95% confidence interval. The answers to parts (d) and (e) were better. In part (e), there was some concern as to the validity of combining two samples. However, as both samples were stated to be random samples the combined sample would also be a random sample.